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Lock ring actuator for tubing hanger installation

Abstract

A hanger assembly includes: a main body; a lock ring coupled to the main body and configured to expand radially outward into a profile on an inner diameter of the wellhead; and an actuator, wherein the actuator is capable of transferring an axial force from the actuator into outward radial expansion of the lock ring toward the profile on the inner diameter of the wellhead. An interface between the actuator and the lock ring is shaped such that: axially downward movement of the actuator from a starting position to an intermediate position actuates the lock ring into the profile; and axially downward movement of the actuator from the intermediate position to a pre-load position applies a pre-load to the hanger assembly.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation-in-part of U.S. application Ser. No. 17/957,921, entitled “RIGIDIZED SEAL ASSEMBLY USING AUTOMATED SPACE-OUT MECHANISM”, filed on Sep. 30, 2022, which is a continuation-in-part of U.S. application Ser. No. 17/118,100, entitled “TUBING HANGER SPACE-OUT MECHANISM”, filed on Dec. 10, 2020, now U.S. Pat. No. 11,459,843, issued Oct. 4, 2022, which claims the benefit of U.S. Provisional Patent Application No. 62/947,506, entitled “TUBING HANGER SPACE-OUT MECHANISM”, filed on Dec. 12, 2019, the entire disclosures of which are incorporated herein by reference.

BACKGROUND

(1) Conventional wellhead systems include a wellhead housing and a subsurface casing string extending from the wellhead into the well bore. During a drilling procedure, a drilling riser and BOP are installed above a wellhead housing (casing head) to provide pressure control as casing is installed, with each casing string having a casing hanger on its upper end for landing on a shoulder within the wellhead housing. Successive casing hangers carrying casing strings of decreasing diameter are installed through the wellbore, and then, a tubing string is installed through the well bore. A tubing hanger connectable to the upper end of the tubing string is supported within the wellhead housing above the last casing hanger, which carries the smallest diameter casing string, for suspending the tubing string within the casing string. Upon completion of this process, the BOP is replaced by a Christmas tree installed above the wellhead housing, with the tree having a valve to enable the oil or gas to be produced and directed into flow lines for transportation to a desired facility.

(2) For various reasons, a tubing hanger or casing hanger within the wellhead may move axially upward, particularly when the wellhead is part of a production system where downhole fluids at elevated temperatures thermally expand the casing string and thus exert a substantial upward force on the casing hanger. Since the casing hanger seal is intended for sealing at a particular location on the wellhead, upward movement of the casing hanger and the seal assembly is detrimental to reliably sealing the casing annulus. Further, for various reasons, the casing hanger may stack higher than intended. Thus, it must be ensured that the tubing hanger is properly sized to lock to the wellhead and that the casing hanger is prevented from moving axially in response to such axial forces.

(3) Various tubing hanger designs and methods have been conceived of for ensuring the tubing hanger is locked to the wellhead housing and the tubing hanger system and casing hanger are rigidized (locked axially) within the wellhead housing. A tubing hanger, once run in and locked into the wellhead, is intended to prevent axial movement of the uppermost casing hanger and seal assembly with respect to the wellhead. Typically, a tubing hanger is run into the wellhead, landed on the casing hanger, and locked to a locking profile on an inner wall of the wellhead housing, which also acts to secure the casing hanger within the wellhead. To install existing tubing hangers, it is first necessary to run a lead impression tool into the wellhead to measure the distance between the top of the casing hanger and the housing locking profile. The lead impression tool is a small block of soft metal, usually lead, which is lowered into the wellhead to take an impression to determine the internal profile of the wellhead, which after being retrieved can be measured to determine the distance between the top of the casing hanger and the housing locking profile. With this information, the tubing hanger can be adjusted at the surface so that once the tubing hanger is run in and secured to the wellhead, it provides a zero-gap connection between the tubing hanger,

the casing hanger, and the wellhead housing and creates any desired pre-load.

(4) This process of taking measurements in the wellhead via a lead impression tool, retrieving the tool to the surface, and then adjusting and installing a tubing hanger into the wellhead is a time-consuming installation process requiring multiple trips into the wellhead. It is now recognized that a need exists for a tubing hanger system that allows for a single-trip installation process.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) For a more complete understanding of the present disclosure and its features and advantages, reference is now made to the following description, taken in conjunction with the accompanying drawings, in which:
- (2) FIG. 1A is a partial cutaway view of a wellhead system having a tubing hanger system, in accordance with an embodiment of the present disclosure;
- (3) FIG. 1B is a close-up of the partial cutaway view of the wellhead system having the tubing hanger system of FIG. 1A, in accordance with an embodiment of the present disclosure;
- (4) FIG. 1C is a close-up of the partial cutaway view of the wellhead system having the tubing hanger system of FIG. 1A, in accordance with an embodiment of the present disclosure;
- (5) FIG. 2 is a partial cutaway view of the tubing hanger system of FIG. 1A, in accordance with an embodiment of the present disclosure;
- (6) FIG. 3A is a cutaway view of a ramp ring and a piston of the tubing hanger system of FIG. 1A where the ramp ring is disposed in an initial position, in accordance with an embodiment of the present disclosure;
- (7) FIG. 3B is a cutaway view of a ramp ring and a piston of the tubing hanger system of FIG. 1A where the ramp ring is disposed in a rotated position, in accordance with an embodiment of the present disclosure;
- (8) FIG. 4 is a partial cross-sectional view of the wellhead system having the tubing hanger system of FIG. 1A, in accordance with an embodiment of the present disclosure.
- (9) FIG. 5A is a top-down cross-sectional view of a tubing hanger system, in accordance with an embodiment of the present disclosure;
- (10) FIG. 5B is a partial perspective view of the tubing hanger system of FIG. 5A, in accordance with an embodiment of the present disclosure;
- (11) FIG. 6A is an isometric view of a tubing hanger system, in accordance with an embodiment of the present disclosure;
- (12) FIG. 6B is a partial top-down cross-sectional view of the tubing hanger system of FIG. 6A, in accordance with an embodiment of the present disclosure;
- (13) FIG. 7 is an isometric view of a ramp ring rotating mechanism, in accordance with an embodiment of the present disclosure;
- (14) FIG. 8A is an isometric view of a ramp ring rotating mechanism, in accordance with an embodiment of the present disclosure;
- (15) FIG. 8B is a partial perspective view of the ramp ring rotating mechanism of FIG. 8A, in accordance with an embodiment of the present disclosure;
- (16) FIG. 9A is an isometric view of a ramp ring rotating mechanism, in accordance with an embodiment of the present disclosure;
- (17) FIG. 9B is a partial perspective view of the ramp ring rotating mechanism of FIG. 9A, in accordance with an embodiment of the present disclosure;
- (18) FIG. 10 is a partial cutaway view of a tubing hanger system, in accordance with an embodiment of the present disclosure;
- (19) FIG. 11 is a partial cross-sectional view of a tubing hanger locking system, in accordance with

an embodiment of the present disclosure; and

(20) FIG. 12 is a partial cross-sectional view of a seal assembly having enhanced rigidity, in accordance with an embodiment of the present disclosure.

(21) FIGS. 13A-13C are cross-sectional views of a system including a tubing hanger with space-out mechanism being run into a wellhead housing and a tapered lock ring actuator, in accordance with an embodiment of the present disclosure.

(22) FIGS. 14A-14C are cross-sectional views of the system of FIGS. 13A-13C as the lock ring is moved out to engage grooves in the wellhead housing, in accordance with an embodiment of the present disclosure.

(23) FIGS. 15A-15C are cross-sectional views of the system of FIGS. 13A-13C as the space-out mechanism actuates, in accordance with an embodiment of the present disclosure.

(24) FIGS. 16A-16C are cross-sectional views of the system of FIGS. 13A-13C as the tapered lock ring actuator moves downward to add pre-load to the tubing hanger, in accordance with an embodiment of the present disclosure.

(25) FIGS. 17A-17D are cross-sectional views of a dual flat lock ring actuator being used during a tubing hanger installation process, in accordance with an embodiment of the present disclosure.

(26) FIGS. 18A-18D are cross-sectional views of a flat to tapered lock ring actuator being used during a tubing hanger installation process, in accordance with an embodiment of the present disclosure.

(27) FIGS. 19A-19D are cross-sectional views of a low angle tapered lock ring actuator being used during a tubing hanger installation process, in accordance with an embodiment of the present disclosure.

DETAILED DESCRIPTION

(28) Illustrative embodiments of the present disclosure are described in detail herein. In the interest of clarity, not all features of an actual implementation are described in this specification. It will of course be appreciated that in the development of any such actual embodiment, numerous implementation specific decisions must be made to achieve developers' specific goals, such as compliance with system related and business related constraints, which will vary from one implementation to another. Moreover, it will be appreciated that such a development effort might be complex and time consuming, but would nevertheless be a routine undertaking for those of ordinary skill in the art having the benefit of the present disclosure. Furthermore, in no way should the following examples be read to limit, or define, the scope of the disclosure.

(29) Certain embodiments of the present disclosure may be directed to a tubing hanger system that may be installed within a wellhead system in a single trip. The tubing hanger system may include multiple pieces that are coupled together such that the tubing hanger may be locked to an inner wall of a high-pressure wellhead housing while applying a preload on a casing hanger, thereby rigidizing the tubing hanger system and casing hanger within the wellhead housing. The tubing hanger system may be run into the wellhead system until the tubing hanger system abuts the casing hanger. Then, the tubing hanger system may be picked up until the tubing hanger system is locked against an inner wall of the high-pressure housing. Lastly, a space-out mechanism of the tubing hanger system may actuate such that it takes up any gaps formed axially by being picked up, thus rigidizing the tubing hanger system and casing hanger within the wellhead housing. The installation process for the tubing hanger system may be accomplished entirely during a single trip into the wellhead as opposed to a first trip with a lead impression tool followed by an adjustment of the tubing hanger system at the surface and a subsequent trip downhole to install the adjusted tubing hanger system. The disclosed systems and method provide both time savings (since only one trip into the wellhead is necessary) and cost savings (since an additional lead impression tool is not required) compared to existing tubing hanger installation techniques.

(30) Certain embodiments of the present disclosure may also be directed to a seal assembly having enhanced rigidity. The seal assembly may be configured such that fluid may apply pressure to the

inner diameter of the seal assembly's lower body, thereby pushing the lower body down. When the lower body is pushed down, a pressure-actuated release mechanism (such as a shear pin) may be actuated (e.g., broken), allowing the lower body to descend further while a ramp ring and spring reduce the size of any existing gap between the casing hanger and wellhead. Such embodiments allow for enhanced rigidization of the wellhead system with minimal cost.

(31) Referring now to FIGS. 1A-3B, certain components of a wellhead system **1** are illustrated according to one or more embodiments of the present disclosure. The illustrated wellhead system **1** may be a subsea wellhead assembly. However, similar techniques may be used in land-based wellhead systems as well. The wellhead system **1** may include a wellhead housing **2**, a casing hanger **10**, a tubing hanger system **20**, and a locking mechanism **60**. The casing hanger **10** may be landed within the wellhead housing **2**. The tubing hanger system **20** may then be landed upon the casing hanger **10** within the wellhead housing **2**. Lastly, the locking mechanism **60** may be landed upon the tubing hanger system **20** within the wellhead housing **2**. The wellhead housing **2** may include a central bore **3** having locking profile **4** disposed thereon. The locking mechanism **60** may engage the locking profile **4** of the wellhead housing **2** in order to lock the casing hanger **10**, the tubing hanger system **20**, and the locking mechanism **60** in place within the wellhead housing **2** and rigidize the system.

(32) The casing hanger **10** may include a casing hanger body **11** having an upper load shoulder **12** and a radially interior profile **13**. The upper load shoulder **12** may be tapered inwards towards the interior profile **13** and ridges may be formed along the upper load shoulder **12**. However, one of ordinary skill in the art would understand that in other embodiments, the upper load shoulder may be tapered outwards away from the interior profile or may not be tapered at all. Additionally, one of ordinary skill in the art would understand that in other embodiments, the upper load shoulder may be smooth or curved instead of having ridges.

(33) The tubing hanger system **20** may include a tubing hanger body **30** and a space-out mechanism **100**. In one or more embodiments, the space-out mechanism may include a ramp ring **40** and a piston **50**. However, one of ordinary skill would understand that space-out mechanisms of other embodiments may include a plurality of ramp rings or wedges. The tubing hanger body **30**, the ramp ring **40**, and the piston **50** may be assembled together before being inserted into the wellhead housing **2** such that the tubing hanger system **20** may be installed in a single trip. The manner in which each of the parts in the tubing hanger system **20** are coupled will be discussed further below. Additionally, the tubing hanger system **20** may be run into the wellhead housing **2** and disposed such that the tubing hanger body **30** seals against the interior profile **13** of the casing hanger body **11** and the piston **50** abuts the upper load shoulder **12** of the casing hanger **10**. In one or more embodiments, to ensure that tubing hanger system **20** is properly seated on the casing hanger **10**, one or more safety lock mechanisms may be used. The safety lock mechanisms according to one or more embodiments of the present disclosure will be discussed further below.

(34) Still referring to FIGS. 1A-3B, the tubing hanger body **30**, according to one or more embodiments of the present disclosure, may include a radially exterior profile **31** defined, in part, by a first sealing profile **32**, a second sealing profile **33**, an upward facing contact surface **34**, a downward facing contact surface **35**, and an axially extending pin slot **36**. The first sealing profile **32** may include a first seal groove **32a** in which a tubing hanger to casing hanger seal **37** is disposed, a second seal groove **32b** in which an o-ring may be disposed, a third groove **32c** in which a retainer ring **21** may be disposed, and a fourth seal groove **32d** in which a first tubing hanger to piston seal **38** may be disposed. The second sealing profile **33** may include groove **33a** in which a second tubing hanger to piston seal **39** may be disposed.

(35) Further, the ramp ring **40** of the space-out mechanism **100**, according to one or more embodiments of the present disclosure, may include an upper contact surface **41**, ramp surfaces **42**, and rotational stop surfaces **43**. The ramp ring **40** may be disposed adjacent to the tubing hanger body **30** such that the ramp ring **40** is positioned about the second sealing surface **33** of the tubing

hanger body **30** and, at least when the tubing hanger system **20** is run-in and when the tubing hanger system **20** is in a fully locked position, the upper contact surface **41** may contact the downward facing contact surface **35** of the tubing hanger body **30**. Additionally, in one or more embodiments, a bottom of the ramp ring **40** may have a plurality of ramp surfaces **42** and a plurality of rotational stop surfaces **43**. By way of example, in one or more embodiments, the ramp ring **40** may include three ramp surfaces each extending 120° circumferentially about the ramp ring **40**. However, one of ordinary skill in the art will understand that in other embodiments, the ramp ring may have a single ramp surface and a single rotational stop surface or any combination of equal numbers of ramp surfaces and rotational stop surfaces that match the number of ramp surfaces and rotational stop surfaces of the piston. Further, in one or more embodiments of the present disclosure, the ramp surfaces **42** may have a constant 3.5° taper. However, one of ordinary skill in the art will understand that in other embodiments the ramp surface may include steps or ridges and/or may have a constant or changing taper in the range of 0.5°-7°. Alternatively, the ramp surface may include any range of angles, surface geometries, and/or coatings that prevent rotation once installed.

(36) Additionally, the piston **50** of the space-out mechanism **100** may include a lower load shoulder **51**, a first interior seal surface **52**, a second interior seal surface **53**, an interior shoulder **54**, ramp surfaces **55**, rotational stop surfaces **56**, and a threaded pin borehole **57**. The piston **50** may be disposed adjacent to the casing hanger **10**, the tubing hanger body **30**, and the ramp ring **40** such that piston is positioned about the first sealing surface **32** and the second sealing surface **33** of the tubing hanger body **30**. Further, the piston **50** may abut the casing hanger **20** on one side and the ramp ring **40** on the other side. Thus, in one or more embodiments, the lower load shoulder **51** may abut the upper load shoulder **12** of the casing hanger **10**. As such, the lower load shoulder **51** may be tapered to match the taper of the upper load shoulder **12** of the casing hanger **10** and ridges may be formed along the lower load shoulder **51** to match the ridges of the upper load shoulder **12** of the casing hanger **10**. However, as discussed above with regard to the upper load shoulder **12** of the casing hanger **10**, one of ordinary skill in the art would understand that in other embodiments, the lower load shoulder may be tapered in a number of ways as long as the taper of the lower load shoulder matches the taper of the upper load shoulder. Additionally, one of ordinary skill in the art would understand that in other embodiments, the upper load shoulder may be smooth or curved instead of having ridges.

(37) Further, the first interior seal surface **52** and second interior seal surface **53** of the piston **50** may be disposed such that when the tubing hanger system **20** is fully assembled, the first tubing hanger to piston seal **38** and the second tubing hanger to piston seal **39** may seal against the first interior seal surface **52** and the second interior seal surface **53** of the piston **50**, respectively. Furthermore, when the tubing hanger system **20** is disposed within the wellhead housing **2** and landed on the casing hanger **10**, the first sealing profile **32** of the tubing hanger body **30** may sit within the casing hanger **10** such that the tubing hanger to casing hanger seal **37** seals against the interior profile **13** of the casing hanger **10**. This sealing profile created between the casing hanger **10**, the tubing hanger body **30**, and the piston **50** may create a piston force that acts in a downward direction against the interior shoulder **54** of the piston **50**, which may hold the piston **50** in abutment with the casing hanger **10** in the event that the tubing hanger body **30** is shifted in an upward direction. Additionally, in one or more embodiments, the threaded pin borehole **57** of the piston **50** may be aligned with the pin slot **36** of the tubing hanger body **30**, and an anti-rotation pin **24** may be coupled to the threaded pin borehole **57** such that the anti-rotation pin **24** rests within the pin slot **36**. This anti-rotation pin, according to one or more embodiments of the present disclosure, may rotationally couple the piston to the tubing hanger body **30** such that the ramp ring **40** may rotate relative to the piston **50** while allowing the tubing hanger body **30** to move axially relative to the piston **50** so that any gap that is formed in locking the tubing hanger system **20** and casing hanger **10** to the wellhead housing **2** may be filled. However, one of ordinary skill in the art

would understand that in other embodiments a variety of methods may be used to rotationally secure the piston and the tubing hanger body such that the ramp ring may rotate relative to the tubing hanger body without also rotating the piston.

(38) Furthermore, still referring to FIGS. 1A-3B, the ramp surfaces **55** of the piston **50** may be configured to abut the ramp surfaces **42** of the ramp ring **40** at least when the tubing hanger system **20** is run-in and when the tubing hanger system **20** is in a fully locked position. As discussed above with regard to the ramp surfaces **42** of the ramp ring **40**, the ramp surfaces **55** of the piston **50** may be designed in various ways so long as the ramp surfaces **55** match the ramp surfaces **42**. By way of example, in one or more embodiments, the piston may include three ramp surfaces each extending 120° circumferentially about the piston **50**. However, one of ordinary skill in the art will understand that in other embodiments, the piston may have a single ramp surface and a single rotational stop surface or any combination of equal numbers of ramp surfaces and rotational stop surfaces that match the number of ramp surfaces and rotational stop surfaces of the ramp ring. Further, in one or more embodiments of the present disclosure, the ramp surfaces **55** may have a constant 3.5° taper. However, one of ordinary skill in the art will understand that in other embodiments the ramp surface may include steps or ridges and/or may have a constant or changing taper in the range of 0.5°-7°. One of ordinary skill will appreciate that the ramp surfaces are designed such that the contact between the ramp surfaces is self-locking and compressive forces between the surfaces will not cause the piston and ramp ring to rotate relative to each other once the tubing hanger system is in the fully locked position. Further, the rotational stop surfaces **56** of the piston **50** and the rotational stop surfaces **43** of the ramp ring **40** may be configured to abut each other at least when the tubing hanger system **20** is run in and may prevent the piston and ramp ring from rotating relative to each other in one direction.

(39) Additionally, the locking mechanism **60**, according to one or more embodiments of the present disclosure, may include a locking mandrel **61** and locking dogs **62**. The plurality of locking dogs **62** may be supported around the locking mandrel **61**. The locking mechanism **60** may be run into the wellhead housing **2** until the locking mechanism **60** abuts the upward facing contact surface **34** of the tubing hanger body **30**. In one or more embodiments, a bottom surface of the locking dogs **62** may directly abut the upward facing contact surface **34** and may be pushed outward into the locking profile **4** of the wellhead housing **2** by a compressive force caused by the locking mandrel **61** pushing down on the locking dogs **62**. The locking dogs **62** may have ridges disposed on an outer surface that match the locking profile **4** disposed along the central bore **3** of the wellhead housing **2**.

(40) Further, the tubing hanger system **20** may include one or more safety locks to ensure that the system is properly run into the wellhead housing **2** and features of the system are not activated prematurely. By way of example, in one or more embodiments, a retainer ring **21** may be included in the tubing hanger system **20** so as to make sure that the piston **50** is properly seated upon the casing hanger **10** and the seals of the tubing hanger body **30** are set within the piston **50** and the casing hanger **10** as necessary for the system to function properly. The retainer ring **21** may be a split ring disposed within the third groove **32c** of the tubing hanger body **30** and may have an uncollapsed outer diameter that is greater than both the diameter of the interior profile **13** of the casing housing **10** and the first interior seal surface **52** of the piston **50**. Further, in a pre-run-in assembled state, the third groove **32c** and the retainer ring **21** may be disposed below the lower load shoulder **51** of the piston **50**. This disposition of the retainer ring **21** and third groove **32c** may be such that the lower load shoulder **51** of the piston **50** cannot abut the upper load shoulder **12** of the casing hanger **10** until the retainer ring **21** is collapsed into the third groove **32c**. The retainer ring **21** may include an upper contact surface **22** and a lower contact surface **23**. The lower contact surface **23** may be tapered such that downward forces from the piston **50** and/or tubing hanger body **30** during run-in push the tapered lower contact surface **23** into an interior edge of the upper load shoulder **12** of the casing hanger **10** and cause the retainer ring **21** to collapse into the third groove

32c. Once collapsed, the outer diameter of the retainer ring 21 may be smaller than the interior profile 13 of the casing hanger 10, allowing the tubing hanger system 20 to properly seat within and against the casing hanger 10. Thus, in one or more embodiments, the retainer ring 21 needs to be collapsed in order for the lower load shoulder 51 of the piston 50 to be able to abut the upper load shoulder 12 of the casing hanger 10. Additionally, various other safety locks may be used in one or more embodiments of the present disclosure.

(41) Referring now to FIG. 4, another safety mechanism according to one or more embodiments of the present disclosure is illustrated. A spring loaded pin disposed within the ramp ring 40 may be installed during assembly of the tubing hanger system 20 and engage the tubing hanger body 30 so as to rotationally lock the ramp ring to the tubing hanger body until the proper time in the tubing hanger system run-in in which the ramp ring must be rotationally actuated in order to take up any axial space created by the installation procedure.

(42) The safety mechanism of the tubing hanger system 20 may include a safety lock pin 70, a safety lock spring 71, and a safety lock rod 72. The safety lock pin 70 and the safety lock spring 71 may be disposed within the ramp ring 40, and the safety lock rod 72 may be disposed within the tubing hanger body 30. The ramp ring 40, in one or more embodiments, may include a pin blind hole 44 disposed in an upper contact surface 41 and a pin securing mechanism 45. The safety lock spring 71 may be disposed within the pin blind hole 44 abutting a bottom of the blind hole, and the safety lock 70 pin may be disposed above the safety lock spring 71 in the blind hole such that the safety lock pin 70 is pushed up towards the tubing hanger body 30. The safety lock pin 70 may include a safety lock pin body 70a and a safety lock pin flange 70b, in which the diameter of the safety lock pin flange 70b is greater than the diameter of the safety lock pin body 70a. The pin securing mechanism 45 may be disposed in the opening of the pin blind hole 44 and may have an inner diameter larger than the safety lock pin body 70a but smaller than the diameter of the safety lock pin flange 70b such that the safety lock pin 70 is maintained within the pin blind hole 44 while the safety lock pin body 70a is able to extend past the upper contact surface 41 of the ramp ring 40.

(43) Additionally, the tubing hanger body 30, in one or more embodiments, may include an elongated hole 58 that extends from an upward facing contact surface 34 to a downward facing contact surface 35. Further, a pin counterbore 59 may be sunk into the downward facing contact surface 35 and concentric with the hole 58. An inner diameter of the pin counterbore 59 may be slightly larger than the outer diameter of the safety lock pin body 70a, and the pin counterbore 59 may be configured to receive the safety lock pin 70 when the tubing hanger system 20 is assembled before run-in. Further, the safety lock rod 72 may be disposed within the hole 58. The safety lock rod 72 may be longer than the length of the hole 58 and the pin counterbore 59 such that when the safety lock pin 70 extends into the pin counterbore 59, the top end 72a of the safety lock rod 72 extends above the upward facing contact surface 34 and when the safety lock rod 72 is compressed down to the upward facing contact surface 34 into the hole 58, the bottom end 72b of the safety lock rod 72 is even with or extends slightly below the downward facing contact surface 35.

(44) Further referring to FIG. 4, in one or more embodiments of the present disclosure, when the tubing hanger system 20 is assembled before run-in, the safety lock pin 70 may engage the pin counterbore 59. During the installation of the tubing hanger system 20 within a wellhead housing, installation of a locking mechanism may cause a locking mandrel to compress the safety lock rod 72 into the hole 58, which will cause the bottom end 72b of the safety lock rod 72 to push the safety lock pin 70 out of the pin counterbore 59. Once the safety lock pin 70 is removed from the pin counterbore 59, the tubing hanger body 30 and the ramp ring 40 will no longer be rotationally locked with respect to each other allowing the ramp ring 40 to rotate relative to the piston 50 along their respective ramp surfaces in order to remove any axial gaps in the tubing hanger system 20 created during the process of locking the tubing hanger system within the wellhead housing.

(45) Referring now to FIGS. 5A and 5B, a tubing hanger system 520, according to one or more embodiments of the present disclosure, is illustrated. As discussed previously, the tubing hanger

system **520** may include a tubing hanger body **530** and a space-out mechanism **500**. Further, the space-out mechanism **500** may include a ramp ring **540** and a piston **550**. Additionally, in one or more embodiments, the ramp ring **540** of the space-out mechanism **500** may be rotationally coupled to the tubing hanger body **530** by a circumferential spring mechanism **580**. The circumferential spring mechanism **580** may be coupled to the ramp ring **540** on a first end and to the tubing hanger body **530** on a second end. The circumferential spring mechanism **580** may include a spring **581**, spring connectors **582**, a transfer block **583**, and bolts **584**. The spring **581** may be disposed within a circumferential groove **531** located on the second sealing profile **533** of the tubing hanger body **530**. The circumferential groove **531** may be disposed between the downward facing contact surface (not shown) of the tubing hanger body **530** and the fourth seal groove (not shown), which is disposed on the second sealing profile **533** of the tubing hanger body **530**. Further, the spring **581** may be directly coupled to the tubing hanger body **530** by a spring connector **582** on a first end of the circumferential groove **531** and may be directly coupled to the transfer block **583** by a spring connector **582** within a distal portion of the circumferential groove **531**. The transfer block **583** may be directly coupled to the ramp ring **540** by bolts **584**.

(46) In one or more embodiments, when assembling the tubing hanger system **520** before run-in, the circumferential spring mechanism **580** may be preloaded such that when a safety mechanism rotationally locking the tubing hanger body **530** and the ramp ring **540** is disengaged, the space-out mechanism **500** self-actuates to rotate the ramp ring **540** against the piston **550** to extend the space-out mechanism **500** axially and remove any axial gaps that have formed during installation of the tubing hanger system **520** into wellhead housing. When the space-out mechanism **500** is actuated, the rotation of the ramp ring will cause the ramp surface of the ramp ring **540** to bear against and rotate against the ramp surface of the piston **530** and extend the space-out mechanism **500** axially.

(47) By way of example, in one or more embodiments, the space-out mechanism **500** may be configured such that the preload puts the spring **581** in tension and releasing the safety mechanism causes the spring **581** to pull the ramp ring **540** causing it to rotate against the piston **550**. However, one of ordinary skill would appreciate that in other embodiments, the spring **581** may be preloaded in compression such that releasing the safety mechanism causes the spring to push the ramp ring **540** causing it to rotate against the piston **550**. Additionally, while a single preloaded spring **581** is illustrated in FIGS. 5A and 5B, one of ordinary skill would appreciate that in other embodiments, there may be multiple springs situated in series or in parallel and preloaded in tension, compression, or torsion so as to rotate a ramp surface of the ramp ring **540** against a ramp surface of the piston **550** causing the space-out mechanism to extend axially and fill in any axial gaps created while rigidizing the tubing hanger system and casing hanger within the wellhead housing.

(48) Referring now to FIGS. 6A and 6B, a tubing hanger system **620**, according to one or more embodiments of the present disclosure, is illustrated. As discussed previously, the tubing hanger system **620** may include a tubing hanger body **630** and a space-out mechanism **600**. Further, the space-out mechanism **600** may include a ramp ring **640** and a piston **650**. Additionally, in one or more embodiments, the space-out mechanism may include a ratchet mechanism **680** disposed inside the tubing hanger system **620** that is configured to allow a user to remotely rotate the ramp ring **640** as necessary during run-in and the process of rigidizing the tubing hanger system **620** within the wellhead housing. The ramp ring **640** may include a plurality of inclined grooves **641** disposed circumferentially along its inner diameter. The ratchet mechanism **680** may be configured to engage the grooves **641** of the ramp ring **640** such that each stroke of the ratchet mechanism rotates the ramp ring **640** by the radial distance of a single groove. The ratchet mechanism **680**, according to one or more embodiments of the present disclosure, may be a short stroke piston with a ratchet. The ratchet mechanism **680** may include a piston **681**, a spring **682**, an actuation arm **683**, and a lever **684**. The piston **681** and the spring **682** may be coaxially disposed with one end of the actuation arm **683** coupled to one end of the piston **681**. Further, the other end of the actuation arm may be coupled to the lever **684**, which is itself pinned to a non-moving portion of the piston **681**,

in order to force the lever **684** to rotate about the pinned connection. The piston **681** may be remotely controlled by a user so as to actuate the ratchet mechanism **680** by pulling the actuation arm **683** such that the lever **684** rotates out of the groove it is sitting in and then allowing the lever **684** to rotate back against the edge of a groove under the force of the spring **682**, which causes the actuation arm to return the lever to its resting position, such that the lever **684** now engages an adjacent groove; thus, rotating the ramp ring **640**, accordingly. Further, as discussed above, rotating the ramp ring **640** causes the ramp ring **640** to shift against the piston **650** to extend the space-out mechanism **600** axially and remove any axial gaps that have formed during installation of the tubing hanger system **620** into the wellhead housing. When the space-out mechanism **600** is actuated, the rotation of the ramp ring will cause the ramp surface of the ramp ring **640** to bear against and rotate against the ramp surface of the piston **630** and extend the space-out mechanism **600** axially.

(49) Referring now to FIG. 7, a ramp ring rotating mechanism **780**, according to one or more embodiments of the present disclosure, is illustrated. A space-out mechanism may include the ramp ring rotating mechanism **780** coupled to a ramp ring. The ramp ring rotating mechanism **780** may include a piston **781** and a curved piston rod **782**. In one or more embodiments, the curved piston rod **782** may be 3-D printed. Further, the curved piston rod **782** may be disposed within the piston **781** and extend from the piston **781**. An end of the curved piston rod **782** may be coupled to the ramp ring, and actuating the piston **781** may cause the curved piston rod **782** to extend, thus causing the ramp ring to rotate relative to a tubing hanger body and a piston **750** of a tubing hanger system. Further, as discussed above, rotating the ramp ring may cause the ramp ring to shift against the piston **750** to extend the space-out mechanism axially and remove any axial gaps that have formed during installation of the tubing hanger system into a wellhead housing. When the space-out mechanism is actuated, the rotation of the ramp ring may cause the ramp surface of the ramp ring to bear against and rotate against the ramp surface of the piston **750** and extend the space-out mechanism axially.

(50) Referring now to FIGS. 8A and 8B, a ramp ring rotating mechanism **880**, according to one or more embodiments of the present disclosure, is illustrated. A space-out mechanism **800** may include the ramp ring rotating mechanism **880** coupled to a ramp ring **840**. The ramp ring rotating mechanism **880** may include a piston **881**, an arm **882**, and a slider **883**. In one or more embodiments, the arm **882** may be coupled to the piston **881** and may be rotated by way of actuation of the piston **881**, which may be operated remotely by a user. An end of the arm **882** may be coupled to a first end of the slider **883**, and a second end of the slider **883** may be coupled to the ramp ring **840**. In one or more embodiments, the slider **883** may be coupled to the arm **882** and the ramp ring **840** by pins. Further, actuating the piston **881** may cause the arm **882** to rotate, thus causing the slider **883** to rotate about the pinned connection to the arm **882** and rotating the ramp ring **840** relative to a tubing hanger body and a piston **850** of a tubing hanger system. Further, as discussed above, rotating the ramp ring **840** may cause the ramp ring **840** to shift against the piston **850** to extend the space-out mechanism **800** axially and remove any axial gaps that have formed during installation of the tubing hanger system into a wellhead housing. When the space-out mechanism **800** is actuated, the rotation of the ramp ring **840** may cause the ramp surface of the ramp ring **840** to bear against and rotate against the ramp surface of the piston **850** and extend the space-out mechanism **800** axially.

(51) Referring now to FIGS. 9A and 9B, a ramp ring rotating mechanism **980**, according to one or more embodiments of the present disclosure, is illustrated. A space-out mechanism **900** may include the ramp ring rotating mechanism **980** coupled to a ramp ring **940**. The ramp ring rotating mechanism **980** may be a geared mechanism and may include a curved rack **981** and a pinion **982**. In one or more embodiments, the curved rack **981** may be coupled to a ramp ring **940** and the pinion **982**. Further, rotation of the pinion **982** may cause rotation of the ramp ring **940** by way of the curved rack **981**, and the pinion **982** may be rotated by remote operation by a user. Therefore, in

one or more embodiments, rotation of the pinion **982** may cause the ramp ring **940** to rotate relative to a tubing hanger body and a piston **950** of a tubing hanger system. Further, as discussed above, rotating the ramp ring **940** may cause the ramp ring **940** to shift against the piston **950** to extend the space-out mechanism **900** axially and remove any axial gaps that have formed during installation of the tubing hanger system into a wellhead housing. When the space-out mechanism **900** is actuated, the rotation of the ramp ring **940** may cause the ramp surface of the ramp ring **940** to bear against and rotate against the ramp surface of the piston **950** and extend the space-out mechanism **900** axially.

(52) Referring now to FIG. **10**, a partial cutaway view of a tubing hanger system **1020**, according to one or more embodiments of the present disclosure, is illustrated. The tubing hanger system **1020** may include a tubing hanger body **1030** and a space-out mechanism **1000**. The space-out mechanism **1000** may include a first ramp ring **1040**, a second ramp ring **1090**, and a piston **1050**. The piston **1050** may include ramp surfaces **1055** and rotational stop surfaces **1056**.

(53) Further, the first ramp ring **1040** may include lower ramp surfaces **1042** and an upper ramp surface **1046**. The lower ramp surfaces **1042** may contact the ramp surfaces **1055** of the piston **1050**, and in one or more embodiments, the ramp surfaces **1042** of the ramp ring **1040** and the ramp surfaces **1055** of the piston **1050** may match in number and taper. By way of example, in one or more embodiments, the ramp ring **1040** may include multiple ramp surfaces **1042** each extending 120° circumferentially about the ramp ring **1040**. However, one of ordinary skill in the art will understand that in other embodiments, the ramp ring may have a single ramp surface and a single rotational stop surface or any combination of equal numbers of ramp surfaces and rotational stop surfaces that match the number of ramp surfaces and rotational stop surfaces of the piston. Further, in one or more embodiments of the present disclosure, the ramp surfaces **1042**, **1055** may all have a constant 4° taper. However, one of ordinary skill in the art will understand that in other embodiments the ramp surface may include steps or ridges and/or may have a constant or changing taper in the range of 0.5°-7°. Additionally, the upper ramp surface **1046** of the first ramp ring **1040** may have a constant taper. In one or more embodiments, the upper ramp surface **1046** may have a constant taper of 0.5°. However, one of ordinary skill in the art will understand that in other embodiments the ramp surface may include steps or ridges and/or may have a constant or changing taper in the range of 0.5°-7°. Further, a pin blind hole **1047** may be formed on the upper ramp surface **1046**.

(54) Furthermore, the second ramp ring **1090** may include a lower ramp surface **1091** and an upper contact surface **1092**. The lower ramp surface **1091** of the second ramp ring **1090** may contact and may match the taper of the upper ramp surface **1046** of the first ramp ring **1040**. As discussed above, the lower ramp surface **1091** may have a constant taper of 0.5°. However, one of ordinary skill in the art will understand that in other embodiments the lower ramp surface **1091** may include steps or ridges and/or may have a constant or changing taper in the range of 0.5°-7° that matches that of the upper ramp surface **1046** of the first ramp ring **1040**. Further, a pin blind hole **1093** may be formed on the lower ramp surface **1091** and may be coaxially aligned with the pin blind hole **1047** of the first ramp ring **1040** during assembly. Further, a shear pin **1095** may be disposed within the aligned pin blind holes **1047**, **1093** to rotationally lock the first ramp ring **1040** and the second ramp ring **1090** until a sufficient piston force is applied to either the first ramp ring **1040** or the second ramp ring **1090** to shear the shear pin **1095** when locking and rigidizing the tubing hanger system **1020** within a wellhead housing.

(55) Additionally, the tubing hanger body **1030** may include a downward facing contact surface **1035**. The downward facing contact surface **1035** of the tubing hanger body **1030** may contact upper contact surface **1092** of the second ramp ring **1090** at least when the tubing hanger system **1020** is run-in and when the tubing hanger system **1020** is in a fully locked and rigidized position within the wellhead housing.

(56) Referring now to FIG. **11**, a tubing hanger locking system **1100**, according to one or more

embodiments of the present disclosure, is illustrated. The tubing hanger locking system **1100** may include, at least, a piston **1110**, locking dogs **1120**, and a wedge **1130**. The piston **1110**, the locking dogs **1120**, and the wedge **1130** may be configured and coupled such that the tubing hanger locking system **1100** locks a tubing hanger in place within a wellhead housing and rigidizes a tubing hanger and casing hanger within the wellhead housing.

(57) While one or more embodiments of the present disclosure may include a piston **50**, **550**, **650**, **750**, **850**, **950**, **1050**, one of ordinary skill would appreciate that in other embodiments, a space-out mechanism of a tubing hanger system may instead include a lower member, which may be a non-actuating member. However, as discussed above with respect to pistons of one or more embodiments of the present disclosure, the lower member may include, at least, ramp surfaces and rotational stop surfaces and may be configured to interact with a ramp ring in order to lock a casing hanger and a tubing hanger system in place within a wellhead housing and rigidize the system.

(58) It should be understood that the present disclosure contemplates a method to lock and rigidize a tubing hanger system and casing hanger within a wellhead housing. The present disclosure also contemplates a method to assemble a tubing hanger system.

(59) In one or more embodiments of the present disclosure, assembly of the tubing hanger system may include disposing a space-out mechanism about a first sealing profile and second sealing profile of a tubing hanger body. Further, in one or more embodiments where the space-out mechanism includes a ramp ring and a piston, a ramp ring may be disposed about the second sealing profile of the tubing hanger body. Then, in one or more embodiments including a safety mechanism for locking a rotation of the ramp ring relative to the tubing hanger body, the portions of the safety mechanism in the ramp ring and in the tubing hanger body may be aligned and coupled. This may further include disposing a safety lock spring in a pin blind hole, disposing a safety lock pin on top of the safety lock spring in the pin blind hole, and disposing a pin securing mechanism into the opening of the pin blind hole. Further, once the safety mechanism for locking a rotation of the ramp ring relative to the tubing hanger body is properly aligned and the safety lock pin is inserted into the pin counterbore of the tubing hanger body, a safety lock rod may be disposed within an elongated hole in the tubing hanger body. Further, if a space-out mechanism requires a pre-load to be applied to a mechanism configured to rotate the ramp ring relative to the tubing hanger body, the pre-load will be applied before rotationally locking the ramp ring and the tubing hanger body by way of the safety mechanism.

(60) Then, in one or more embodiments, a piston may be disposed about the first sealing profile and the second sealing profile of the tubing hanger body. Once the piston is properly installed such that the seals of the tubing hanger body are properly located within the piston, the piston and the tubing hanger body may be aligned such that the anti-rotation pin may be threaded into the threaded pin borehole of the piston and extend into a pin slot of the tubing hanger body. Additionally, in one or more embodiments, a retainer ring may be disposed within a third groove of the tubing hanger body.

(61) Additionally, in one or more embodiments of the present disclosure, locking and rigidizing a tubing hanger system and casing hanger within a wellhead housing may include running an assembled tubing hanger system into the wellhead housing, landing the tubing hanger system on the casing hanger and sealing a tubing hanger to casing hanger seal of the tubing hanger body against the casing hanger. Landing the tubing hanger system on the casing hanger may further include collapsing a retaining ring into a third groove of the tubing hanger body. Then, in one or more embodiments, a seal test on the tubing hanger to casing hanger seal may be performed. Once the seal test confirms that the seals are properly set, the tubing hanger may be locked. The process of locking the tubing hanger may activate the safety lock rod and engage the locking dogs into their locking profile within the wellhead housing. Then, the tubing hanger body may be lifted to preload the locking mechanism in place within the wellhead housing.

(62) In one or more embodiments, the space-out mechanism may then be actuated, taking up any

axial gaps created by lifting on the tubing hanger body and rigidizing the tubing hanger system within the wellhead housing. Actuating the space-out mechanism may further include unlocking a safety mechanism. Unlocking the safety mechanism may include compressing a safety locking rod into an elongated hole of the tubing hanger body and pushing a safety lock pin out of a pin counterbore of the tubing hanger body such that the ramp ring is no longer rotationally locked to the tubing hanger body. Actuating the space-out mechanism may further include moving the piston down to push against the casing hanger, rotating the ramp ring, and filling the gap between the piston and the tubing hanger body. Once the space-out mechanism has been activated to rigidize the tubing hanger body and the casing body within the wellhead housing, the casing hanger seal may be seal tested to ensure that it is still properly sealing. Then, finally, the tubing hanger system may be released.

(63) Space-out mechanisms, as described at length above, may be used in other contexts as well to rigidize wellhead system components by removing any axial gaps in the wellhead system created during the process of landing/locking components of the wellhead system. For example, a space-out mechanism may be used to close out any axial gaps in a connection between a casing hanger and the wellhead housing (e.g., prior to landing a tubing hanger).

(64) In some cases, machining tolerances may give rise to small gaps between a locking mechanism (e.g., lock ring) of a seal assembly and an upper edge of a complementary lock profile of the wellhead when the seal assembly is landed and locked to seal an annulus between the casing hanger and the wellhead. Such gaps may enable the seal assembly located between the casing hanger and the wellhead to move up and down axially in response to pressure differentials. Over time, this motion of the seal may cause undesirable wear on the seal, increasing the chance for failure.

(65) To address this issue, a seal assembly **1200**, as depicted in FIG. 12, may be equipped with a space-out mechanism **1222** used to reduce gaps and rigidize the system. The seal assembly **1200** may include an upper body **1206**, a lower body **1214**, an actuator sleeve **1220**, and a locking mechanism **1216**. The seal assembly **1200** may first be lowered into a wellhead. Once lowered, the seal assembly **1220** may land upon the casing hanger **1224**. Weight may be applied to the seal assembly **1220** via a running tool (not shown); this weight may cause the actuator sleeve **1220** to move downward with respect to the upper body **1206** and the locking mechanism **1216**, thereby pushing the locking mechanism **1216** into grooves of the inner diameter profile of the wellhead housing (not shown). Once the actuator sleeve **1220** has moved down, there may remain a gap between the locking mechanism **1216** and the uppermost edge of the grooves. To increase the rigidity of the system, it is desirable to reduce, minimize, or eliminate such a gap.

(66) The seal assembly **1200** of FIG. 12 includes a space-out mechanism **1222**, which may be coupled between the upper body **1206** and the lower body **1214** as shown. The upper body **1206** may be coupled to the locking mechanism **1216**. The lower body **1214** may land on a landing shoulder of the casing hanger **1224**. The lower body may include a seal **1218**, which may be rigidized in accordance with teachings of the present disclosure. The space-out mechanism **1222** may be utilized to increase the axial distance between the upper body **1206** and the lower body **1214** in order to ensure that (1) the seal **1218** has landed at the appropriate location; and (2) the locking mechanism **1216** is positioned against the upper edge of the corresponding profile of the inner diameter of the wellhead (not shown).

(67) The space-out mechanism **1222** may include a ramp ring **1202** and an actuation mechanism **1203**. The actuation mechanism **1203**, in the illustrated embodiment, includes a spring and **1210** and a pressure-actuated release mechanism **1212**. The pressure-actuated release mechanism **1212** may be any mechanism suitable to prevent the lower body from descending until a threshold pressure is reached; for example, the pressure-actuated release mechanism **1212** may be a shear pin or other shearable actuation component.

(68) In certain embodiments, the ramp ring **1202** may be configured to rotate relative to the upper

body **1206** and the lower body **1214**. The ramp ring **1202** may comprise at least one tapered surface, and the upper body **1206** may comprise at least one tapered surface configured to interface with the at least one tapered surface of the ramp ring **1202**. The tapered surfaces of the ramp ring **1202** and the upper body **1206** may be complementary. Furthermore, the tapered surfaces of the ramp ring **1202** and the upper body **1206** may be configured to bear against each other to rigidize the system. In certain embodiments, the at least one taper of each of the ramp ring **1202** and the upper body **1206** may have a slope between 0.5° and 7° . The ramp ring **1202** of the space-out mechanism **1222**, according to one or more embodiments of the present disclosure, may include a lower contact surface configured to interface with an upper contact surface of the lower body **1214**. Additionally, in one or more embodiments, an upper surface of the ramp ring **1202** may have a plurality of ramp surfaces and a plurality of rotational stop surfaces. For example, the ramp ring **1202** may take the form of the ramp ring (e.g., ramp ring **40**) shown in FIG. 3A, except that the ramp surfaces and stop surfaces are on the top of the ramp ring instead of the bottom. For example, in one or more embodiments, the ramp ring **1202** may include three ramp surfaces each extending 120° circumferentially about the ramp ring **1202**. However, one of ordinary skill in the art will understand that in other embodiments, the ramp ring **1202** may have a single ramp surface and a single rotational stop surface or any combination of equal numbers of ramp surfaces and rotational stop surfaces that match the number of ramp surfaces and rotational stop surfaces of the upper body **1206**. Further, in one or more embodiments of the present disclosure, the ramp surface(s) may have a constant 3.5° taper. However, one of ordinary skill in the art will understand that in other embodiments the ramp surface(s) may include steps or ridges and/or may have a constant or changing taper in the range of 0.5° - 7° . Alternatively, the ramp surface(s) may include any range of angles, surface geometries, and/or coatings that prevent rotation once installed.

(69) The ramp ring **1202** of the space-out mechanism **1222** may be configured to rotate via the spring **1210** relative to the upper body **1206** and the lower body **1214**. In one or more embodiments, the ramp ring **1202** may be rotationally coupled to the lower body **1214** by a circumferential spring mechanism (which may or may not be similar to the circumferential spring mechanism **580** of FIG. 5A) including the spring **1210**. The circumferential spring mechanism may be coupled to the ramp ring **1202** on a first end and to the lower body **1214** on a second end. The spring **1210** may be disposed within a circumferential groove (not shown) located on the lower body **1214**. Further, the spring **1210** may be directly coupled to the lower body **1214** by a spring connector (not shown) on a first end of the circumferential groove and may be coupled (e.g., via a transfer block, spring connector, and/or bolts) to the ramp ring **1202**.

(70) In one or more embodiments, when assembling the seal assembly **1200** before run-in, the circumferential spring mechanism may be preloaded such that when a safety mechanism locking the upper body **1206** to the lower body is disengaged, the space-out mechanism **1222** self-actuates to rotate the ramp ring **1202** against the lower body **1214** to extend the space-out mechanism **1222** axially and remove any axial gaps that have formed during installation of the seal assembly **1200** into wellhead housing. When the space-out mechanism **1222** is actuated, the rotation of the ramp ring **1202** will cause the ramp surface(s) of the ramp ring **1202** to bear against and rotate against the corresponding ramp surface(s) of the upper body **1206** and extend the space-out mechanism **1222** axially.

(71) The pressure-actuated release mechanism **1212** may be a safety mechanism configured to lock the upper body **1206** to the lower body **1214** until a pressure is applied to disengage the pressure-actuated mechanism **1212**. The pressure-actuated release mechanism **1212** may be any mechanism suitable to prevent the lower body **1214** from moving with respect to the upper body **1206** until a threshold pressure is reached; for example, the pressure-actuated release mechanism **1212** may be a shear pin. The shear pin may extend between the upper body **1206** and the lower body **1214**. The actuation mechanism **1203** of the illustrated embodiment including the spring **1210** and shear pin are purely exemplary—any actuation mechanism for the space-out mechanism **1222** may be used

without departing from the scope of the present disclosure. For example, and without limitation, one or more of a spring **1210**, shear pin, other pressure-actuated release mechanism **1212**, piston, ratchet, rod, spring plate, arm, slider, rack, and pinion may be used for actuation, such as those described at length above with reference to FIGS. **6A-10**.

(72) In certain embodiments, the space-out mechanism **1222** may include another safety mechanism according to one or more embodiments of the present disclosure. In particular, a spring-loaded rod **1204** disposed within the ramp ring **40** may be installed during assembly of the seal assembly **1200** and engage the upper body **1206** so as to rotationally lock the ramp ring **1202** to the upper body **1206** until the proper time in the seal assembly run-in sequence in which the ramp ring **1202** should be rotationally actuated in order to take up any axial space created by the installation procedure.

(73) The safety mechanism may be similar to the safety mechanism illustrated in detail in FIG. **4** and described above. The only features of the safety mechanism illustrated in FIG. **12** are the spring-loaded rod **1204** (e.g., safety lock rod **72** of FIG. **4**) and the corresponding pin securing mechanism **1208** (e.g., pin securing mechanism **45** of FIG. **4**). It will be understood that the safety mechanism of FIG. **12** may also include a safety lock pin (e.g., **70** of FIG. **4**) and a safety lock spring (e.g., **71** of FIG. **4**) disposed within the ramp ring **1202**. The spring-loaded rod **1204** may be disposed within the upper body **1206**. The ramp ring **1202**, in one or more embodiments, may include a pin blind hole disposed in an upper contact surface thereof with the pin securing mechanism **1208** disposed in the opening of the pin blind hole. Additionally, the upper body **1206**, in one or more embodiments, may include an elongated hole **1209** that extends from an upward facing contact surface thereof to a downward facing contact surface thereof. Further, a pin counterbore (not shown) may be sunk into the downward facing contact surface and concentric with the hole **1209**. The spring-loaded rod **1204** may be disposed within the hole **1209**.

(74) The safety mechanism may function similar to the assembly illustrated in FIG. **4** and described above with reference to FIG. **4**. For example, in one or more embodiments of the present disclosure, when the seal assembly **1200** is assembled before run-in, the safety lock pin (not shown) may engage the pin counterbore (not shown). During the installation of the seal assembly **1200** within the wellhead housing, actuation of the locking mechanism **1216** may cause the actuator sleeve **1220** to compress the spring-loaded rod **1204** into the hole **1209**, which will cause the bottom end of the spring-loaded rod **1204** to push the safety lock pin out of the pin counterbore. Once the safety lock pin is removed from the pin counterbore, the upper body **1206** and the ramp ring **1202** will no longer be rotationally locked with respect to each other, thereby allowing the ramp ring **1202** to rotate relative to the upper body **1206** (along the respective ramp surfaces) in order to remove any axial gaps in the seal assembly **1200** created during the process of locking and sealing the casing hanger **1224** within the wellhead housing.

(75) The illustrated assembly of FIG. **12** may be utilized in a subsea wellhead system. In certain embodiments, the assembly of FIG. **12** may be utilized by performing a sequence of steps—one exemplary sequence is provided below.

(76) First, the seal assembly **1200** may be landed on the casing hanger **1224** in the wellhead housing (not shown). The actuator sleeve **1220** may then be engaged, thereby extending the locking mechanism in a radially outward direction to lock the seal assembly **1200** to the wellhead and releasing the spring-loaded rod **1204** so that the ramp ring **1202** and the upper body **1206** are rotationally uncoupled. The seal assembly **1200** may then be pulled up via a seal assembly running tool (not shown).

(77) Once the seal assembly **1200** is raised up such that the locking mechanism **1216** is engaged against the top of the inner profile on the wellhead, there may be a gap present between the lower body **1214** and the landing shoulder at the top of the casing hanger **1224**. Accordingly, pressurized fluid may be pumped down an annulus between the outer diameter of the stem of the running tool and the inner diameter of the upper body **1206** and lower body **1214**. The pressurized fluid may

apply pressure to the inner diameter of the seal **1218** at the bottom of the lower body **1214**, thereby causing the lower body **1214** to begin to descend. The lower body's movement may actuate the pressure-actuated release mechanism **1212** (e.g., shearing a shear pin), allowing the lower body **1214** to further descend with respect to the upper body **1206**. Once the lower body **1214** has descended, the ramp ring **1202** and spring **1210** may automatically facilitate the closing of the gap. (78) Finally, the casing hanger/seal assembly running tool may be retrieved. The above list of steps should be understood as non-limiting—additional steps may be added or removed without departing from the scope of the present disclosure. Moreover, steps may be executed in a different order without departing from the scope of the present disclosure.

(79) Locking mechanisms, as already explained in detail above, may be used to lock the tubing hanger assembly to the inner wall of the wellhead housing. Alternatively, in certain embodiments, locking mechanisms may also be used to lock the casing hanger assembly to the inner wall of the wellhead housing. When using a traditional type of locking system with a tubing hanger or casing hanger that has an automatic space-out mechanism, significant pre-load may not be able to be developed on the hanger assembly, e.g., from pressure from below. For example, significant pre-load may be about 100,000 lbs, about 1,000,000 lbs, or any other suitable value.

(80) Referring now to FIGS. **13A-16C**, a locking mechanism **1320** may be employed to address this issue, i.e., to lock a hanger assembly **1300** against a wellhead housing **1302** as well as preload the hanger assembly **1300** with a desired level of pre-load, for example, so as to enhance rigidization of the wellhead system. The hanger assembly **1300** may include an upper body **1304**, a main body **1306**, and a lower body **1308**. The lower body **1308** may first be landed within the wellhead housing **1302**. The main body **1306** may then be landed upon the lower body **1308** within the wellhead housing **1302**. The locking mechanism **1320** may be landed upon the main body **1306** within the wellhead housing **1302**. Lastly, the upper body **1304** may be coupled to the main body **1306** within the wellhead housing **1302**. The wellhead housing **1302** may include a central bore **1310** having a locking profile **1312** disposed thereon. The locking mechanism **1320** may engage the locking profile **1312** of the wellhead housing **1302** in order to lock the hanger assembly **1300** in place within the wellhead housing **1302** and rigidize the system. In one or more embodiments, the locking mechanism **1320** may include a lock ring **1322** and an actuator **1324**. The locking mechanism **1320** may be run into the wellhead housing **1302** until the locking mechanism **1320** abuts an upward-facing contact surface **1314** of the main body **1306**. In one or more embodiments, a bottom surface of the lock ring **1322** may directly abut the upward-facing contact surface **1314** and may be pushed outward in a radial direction into the locking profile **1312** of the wellhead housing **1302** by a compressive force caused by the actuator **1324** pushing down axially on the lock ring **1322**. In particular, the lock ring **1322** may have ridges **1326** disposed on its outer surface that may match the locking profile **1312** disposed along the central bore **1310** of the wellhead housing **1302**.

(81) The lock ring **1322** may be actuated by the actuator **1324** via an interface between the lock ring **1322** and the actuator **1324**. The interface, according to one or more embodiments of the present disclosure, may allow axial force from the actuator **1324** to be transferred through the interface into outward radial expansion of the lock ring **1322** toward the locking profile **1312** on the inner diameter of the wellhead housing **1302**. In particular embodiments, the interface may be designed to achieve a so-called two-stage locking. For example, the interface may be shaped such that (1) axially downward movement of the actuator **1324** from a starting position to an intermediate position may actuate the lock ring **1322** into the locking profile **1312**; and (2) axially downward movement of the actuator **1324** from the intermediate position to a pre-load position may apply a pre-load to the hanger assembly **1300**, the details of which will be discussed further below.

(82) In the illustrated embodiment, the interface may include a single tapered surface **1328** of the actuator **1324** that is angled from vertical such that an upper portion of the actuator **1324** may have

a greater diameter than a lower portion of the actuator **1324**. As an example and not by way of limitation, the single tapered surface **1328** may be angled less than 4 degrees from vertical. As another example and not by way of limitation, the single tapered surface **1328** may be angled less than 2 degrees from vertical. Alternatively, one of ordinary skill in the art would understand that in other embodiments, the single tapered surface **1328** may be angled at any suitable angle from vertical without departing from the scope of the present disclosure. In the illustrated embodiment, the interface may also include an inner surface **1330** of the lock ring **1322** that is tapered to match the single tapered surface **1328** of the actuator **1324**. When the actuator **1324** moves down in the axial direction, the single tapered surface **1328** may contact and press against the inner surface **1330** of the lock ring **1322**, pushing the lock ring **1322** radially outward (by a horizontal component of the pressing force acting through the interface) until the lock ring **1322** engages into the locking profile **1312**.

(83) In one or more embodiments, axial downward movement of the actuator **1324** may be interrupted by a split ring **1332** such that the actuator **1324** is temporarily stopped at its intermediate position. In the illustrated embodiment, the split ring **1332** may be disposed along a radially inner surface of the upper body **1304**. For example, the split ring **1332** may be disposed within a groove **1316** of the upper body **1304** and may have an unretracted outer diameter that is greater than the diameter of the upper body **1304**. Axial location of the split ring **1332** may be disposed below the actuator **1324** in its starting state. This position of the split ring **1332** may be such that the upper portion of the actuator **1324** cannot contact the lock ring **1322** until the split ring **1332** is collapsed into the groove **1316**. Further, the split ring **1332** may have an upper contact surface that is tapered such that additional downward forces from the actuator **1324** as it travels past the intermediate position during actuation may push on the upper contact surface and cause the split ring **1332** to retract into the groove **1316**. In the retracted state, the radially outer surface of the split ring **1332** may stay flush with or short of the surface of the upper body **1304**, thereby opening a passage for the actuator **1324** to press down further, e.g., past the split ring **1332** to the pre-load position. Thus, in one or more embodiments, the split ring **1332** needs to be retracted or collapsed in order for the actuator **1324** to be able to travel to the pre-load position. Although the present disclosure describes a hanger assembly with a split ring in a particular manner, the present disclosure contemplates hanger assemblies with any suitable split rings in any suitable manner. For example, additional or alternative to the split ring **1332** disposed in the upper body **1304**, a split ring may be arranged in a running tool (not shown), which serves to apply weight to the actuator **1324**, causing the actuator **1324** to move downward. In particular, the split ring may stay in its unretracted state under low pressure, causing the actuator **1324** to stop at the intermediate position, and may yield as more weight in the running tool is applied, thereby allowing the actuator **1324** to move past the intermediate position when pre-loading the hanger assembly **1300** within the wellhead housing **1302**.

(84) In certain embodiments, the hanger assembly **1300** may include another stopper, e.g., for temporarily stopping the actuator at its intermediate position according to one or more embodiments of the present disclosure. As an example, a shear pin may be disposed at the radially inner surface of the upper body **1304** to block travel path of the actuator **1324** at the intermediate position until a sufficient actuation force is applied to further press down the actuator **1324** and shear the shear pin when pre-loading the hanger assembly **1300** within the wellhead housing **1302**. As another example, a shear pin may be disposed in a running tool (not shown), which serves to apply weight to the actuator **1324**, causing the actuator **1324** to move downward. In particular, the shear pin may prevent movement of the actuator **1324** past the intermediate position until additional weight is added in the running tool to shear the shear pin when pre-loading the hanger assembly **1300** within the wellhead housing **1302**. It should be understood that the shear pins described above are purely exemplary. Any suitable stopper feature may be employed without departing from the scope of the present disclosure. For example, one or more of a lock, a spring, a

ring, a pin, a ratchet, or the like may be employed additionally or alternatively to temporarily retain the actuator **1324** at the intermediate position.

(85) The hanger assembly **1300** as illustrated may include a space-out mechanism **1340** disposed between the main body **1306** and the lower body **1308**, for example, to close out axial gaps in the wellhead and rigidize the system. In one or more embodiments, the space-out mechanism **1340** may include a ramp ring **1342**, which may be configured to rotate relative to the main body **1306** and the lower body **1308**, and a piston **1344**, which may interface the ramp ring **1342**. Specifically, the ramp ring **1342** may comprise at least one tapered surface **1346**, and the piston **1344** may comprise at least one tapered surface configured to interface with the at least one tapered surface **1346** of the ramp ring **1342**. The tapered surfaces of the ramp ring **1342** and the piston **1344** may be complementary. Furthermore, the tapered surfaces of the ramp ring **1342** and the piston **1344** may be configured to bear against each other to rigidize the system. In certain embodiments, the at least one taper of each of the ramp ring **1342** and the piston **1344** may have a slope between 0.5° and 7° . Additionally, in one or more embodiments, the ramp ring **1342** and the piston **1344** may take form similar to the ramp ring illustrated in detail in FIG. 3A and described above with reference to FIG. 3A. When actuated, the ramp ring **1342** may rotate out, thereby expanding the space-out mechanism **1340** to remove any axial gaps in the wellhead system created during installation. In certain embodiments, the ramp ring **1342** may function similar to the ramp ring illustrated in detail in FIG. 3A and described above with reference to FIG. 3A. Alternatively, the ramp ring **1342** may function similar to the ramp ring illustrated in detail in FIG. 12 and described above with reference to FIG. 12.

(86) In certain embodiments, the hanger assembly **1300** may include a space-out mechanism release assembly **1350** configured to selectively release the space-out mechanism **1340** so that the space-out mechanism **1340** can actuate. In particular, a lever-actuated pin **1352** disposed within the main body **1306** may be installed during assembly of the hanger assembly **1300** and engage the ramp ring **1342** so as to rotationally lock the ramp ring **1342** to the main body **1306** until the proper time in the hanger assembly **1300** run-in sequence in which the ramp ring **1342** should be rotationally actuated in order to take up any axial space created by the installation procedure. Specifically, the lever-actuated pin **1352** may be levered to retract up and out of engagement with the ramp ring **1342** by a trigger rod **1354**. In one or more embodiments, one end of the trigger rod **1354** may be coupled to the lever-actuated pin **1352** while the other end of the trigger rod **1354** may be coupled to the actuator **1324**. Upon the actuator **1324** descending from the starting position to the intermediate position, the trigger rod **1354** may be pushed downward to lift up the lever-actuated pin **1352**, thereby unlocking the ramp ring **1342** such that the ramp ring **1342** may be rotationally actuated as needed.

(87) Alternatively, the space-out mechanism release assembly **1350** may be configured similar to the safety mechanism illustrated in detail in FIG. 4 and described above with reference to FIG. 4. For example, in certain embodiments, the space-out mechanism release assembly **1350** may be spring-loaded, pressure-actuated, or movable in any suitable manner without departing from the scope of the present disclosure in order to selectively retain or release the space-out mechanism **1340**.

(88) The embodiments illustrated in FIGS. 13A through 16C may be utilized in a subsea wellhead system. However, similar techniques may be used in land-based wellhead assemblies as well. In certain embodiments, the assembly of FIGS. 13A-16C may be utilized by performing a sequence of steps—one exemplary sequence is provided below.

(89) First, the hanger assembly **1300** may be landed into and engaged with the wellhead housing **1302**. Once the hanger assembly **1300** is in position, the running tool may be actuated with low pressure, causing the actuator **1324** to move downward with respect to the main body **1306** and the lock ring **1322**, thereby expanding the lock ring **1322** radially towards and into the locking profile **1312** of the wellhead housing **1302**. In particular, the actuator **1324** may move from the starting

position until stopped at the intermediate position by the split ring **1332**. Then, the space-out mechanism **1340** may be released for actuation. In one or more embodiments, the release of the space-out mechanism **1340** may occur in response to the actuator **1324** moving from the starting position to the intermediate position. For example, as the actuator **1324** descends, the trigger rod **1354** coupled to the actuator **1324** may push on the lever-actuated pin **1352**, thereby unlocking the space-out mechanism **1340** by rotationally uncoupling the ramp ring **1342** and the main body **1306** so that the space-out mechanism **1340** may actuate as needed. Overpull may be applied on the running tool to pull up the hanger assembly **1300**. In some embodiments, the overpull may pull about 50,000 lbs of tension. Of course, other suitable levels of overpull may be applied without departing from the scope of the present disclosure. Lifting the hanger assembly **1300** in the vertical direction may cause the upper surface of the lock ring **1322** to engage on the locking profile **1312** (e.g., on a topmost lockdown groove of the locking profile **1312**) of the wellhead housing **1302**. This may confirm that the hanger assembly **1300** is locked into the wellhead housing **1302** and the space-out mechanism **1340** will activate (e.g., via the piston **1344**). Then, the overpull may be maintained to retain the hanger assembly **1300** at the lifted position. Pressure is exerted to seal the main body **1306**. In certain embodiments, this may be performed in a way similar to the method described above with reference to FIG. **12**. Upon sealing, the piston **1344** may be pressurized, thereby maintaining the space-out mechanism **1340** in its actuated position, thus ensuring proper space-out. Following confirmation that the main body **1306** is properly sealed and spaced out, pre-load may be applied to the hanger assembly **1300**. In certain embodiments, the lock ring **1322** may be subjected to a pressure of about 15,000 psi or other suitable levels of pressure. This will expand the lock ring **1322** further into the locking profile **1312** of the wellhead housing **1302** and add significant pre-load to the hanger assembly **1300**. In certain embodiments, the pre-load on the hanger assembly **1300** may be up to 100,000 lbs. Of course, other suitable levels of pre-load on the hanger assembly **1300** are also contemplated without departing from the scope of the present disclosure. Finally, the overpull may be removed, and the running tool may be unlatched and retrieved. The above list of steps should be understood as non-limiting—additional steps may be added or removed without departing from the scope of the present disclosure. Moreover, steps may be executed in a different order without departing from the scope of the present disclosure.

(90) Referring now to FIGS. **17A-17D**, interaction of a lock ring **1722** and an actuator **1724**, according to one or more embodiments of the present disclosure, during installation is illustrated. The installation may be performed following the steps and sequence described with reference to FIGS. **13A**, **14A**, **15A**, and **16A**. Alternatively, as will be appreciated by one of skill in the art, the installation may be performed using less or more steps or executed in a different sequence without departing from the scope of the present disclosure. In the illustrated embodiments, the interaction of the lock ring **1722** and the actuator **1724** may occur via an interface **1700**. The interface **1700** may include two flat surfaces **1726**, **1728** of the actuator **1724** that are separated by a step **1730**. For example, the two flat surfaces **1726**, **1728**, and the step **1730** may be formed on the outer surface of the actuator **1724** such that an upper portion of the actuator **1724** may have a greater diameter than a lower portion of the actuator **1724**. While described as being flat, the two surfaces **1726**, **1728** in some embodiments may alternatively be angled, tapered, or otherwise inclined from vertical if needed. Furthermore, the step **1730** may be slightly tapered as shown, or alternatively form an angle of about 90 degrees relative to the vertical. In one or more embodiments, the lock ring **1722** may have an inner surface that is shaped to match the outer surface of the actuator **1724**. As a non-limiting example, the lock ring **1722** may also be structured with two flat surfaces separated by a step. During installation, at the starting position of the actuator **1724**, the actuator **1724** may be spaced from contact the lock ring **1722**. Actuating the actuator **1724** (e.g., under low pressure) may first cause the flat surface **1726** to contact the lock ring **1722** until the step **1730** catches an upper rim of the lock ring **1722** when the actuator **1724** moves from the starting position to the intermediate position. As the actuator **1724** continues to press down on the lock ring **1722**

and moves from the intermediate position to the pre-load position (e.g., following the overpull and actuation of a space-out mechanism as described at length above), the flat surface **1728** may push against the lock ring **1722** to further expand the lock ring **1722** radially into the lock profile **1712**, thereby adding a pre-load to the system.

(91) Referring now to FIGS. **18A-18D**, interaction of a lock ring **1822** and an actuator **1824**, according to one or more embodiments of the present disclosure, during installation is illustrated. The installation may be performed following the steps and sequence described with reference to FIGS. **13A**, **14A**, **15A**, and **16A**. Alternatively, as will be appreciated by one of skill in the art, the installation may be performed using less or more steps or executed in a different sequence without departing from the scope of the present disclosure. In the illustrated embodiments, the interaction of the lock ring **1822** and the actuator **1824** may occur via an interface **1800**. The interface **1800** may include a flat surface **1826** of the actuator **1824** and a tapered surface **1828** of the actuator **1824**. The flat surface **1826** may be located vertically below the tapered surface **1828** such that an upper portion of the actuator **1824** may have a greater diameter than a lower portion of the actuator **1824**. The flat surface **1826** and the tapered surface **1828** may be continuously connected or otherwise separated by a step (not shown). In one or more embodiments, the lock ring **1822** may have an inner surface that is shaped to match the outer surface of the actuator **1824**. As a non-limiting example, the lock ring **1822** may also be structured with a flat surface and a tapered surface. During installation, at the starting position of the actuator **1824**, the actuator **1824** may be spaced from contact the lock ring **1822**. Actuating the actuator **1824** (e.g., under low pressure) may first cause the flat surface **1826** to contact the lock ring **1822** until the connection between the flat surface **1826** and the tapered surface **1828** contacts an upper rim of the lock ring **1822** when the actuator **1824** moves from the starting position to the intermediate position. As the actuator **1824** continues to press down on the lock ring **1822** and moves from the intermediate position to the pre-load position (e.g., following the overpull and actuation of a space-out mechanism as described at length above), the tapered surface **1828** may push against the lock ring **1822** to further expand the lock ring **1822** radially into the lock profile **1812**, thereby adding a pre-load to the system.

(92) Referring now to FIGS. **19A-19D**, interaction of a lock ring **1922** and an actuator **1924**, according to one or more embodiments of the present disclosure, during installation is illustrated. The installation may be performed following the steps and sequence described with reference to FIGS. **13A**, **14A**, **15A**, and **16A**. Alternatively, as will be appreciated by one of skill in the art, the installation may be performed using less or more steps or executed in a different sequence without departing from the scope of the present disclosure. In the illustrated embodiments, the interaction of the lock ring **1922** and the actuator **1924** may occur via an interface **1900**. The interface **1900** may include a tapered surface **1926** of the actuator **1924**. The tapered surface **1926** may be sloped from vertical at a low angle such that an upper portion of the actuator **1924** may have a slightly greater diameter than a lower portion of the actuator **1924**. As an example and not by way of limitation, the tapered surface **1926** may be angled less than 4 degrees from vertical. As another example and not by way of limitation, the tapered surface **1926** may be angled less than 2 degrees from vertical. Alternatively, one of ordinary skill in the art would understand that in other embodiments, the single tapered surface **1926** may be angled at any suitable angle from vertical without departing from the scope of the present disclosure. In one or more embodiments, the lock ring **1922** may have an inner surface that is shaped to match the outer surface of the actuator **1924**. As a non-limiting example, the lock ring **1922** may also be structured with a tapered surface that is sloped from vertical at a low angle. During installation, at the starting position of the actuator **1924**, the actuator **1924** may be spaced from contact the lock ring **1922**. Actuating the actuator **1924** from the starting position to the intermediate position as well as from the intermediate position to the pre-load position may cause the tapered surface **1926** to contact the lock ring **1922** to expand the lock ring **1922** radially into the lock profile **1912**.

(93) It should be understood that the present disclosure contemplates a method to lock and rigidize

a seal assembly and casing hanger within a wellhead housing. The present disclosure also contemplates a method to assemble a seal assembly.

(94) In one or more embodiments of the present disclosure, assembling the seal assembly may include disposing a space-out mechanism between an upper body and a lower body of the seal assembly. Then, in one or more embodiments including a safety mechanism for locking a rotation of the ramp ring relative to the upper body, the portions of the safety mechanism in the ramp ring and in the upper body may be aligned and coupled. This may further include disposing a safety lock spring in a pin blind hole, disposing a safety lock pin on top of the safety lock spring in the pin blind hole, and disposing a pin securing mechanism into the opening of the pin blind hole. Further, once the safety mechanism for locking a rotation of the ramp ring relative to the upper body is properly aligned and the safety lock pin is inserted into the pin counterbore of the tubing hanger body, a spring-loaded rod may be disposed within an elongated hole in the upper body. Further, if a space-out mechanism requires a pre-load to be applied to a mechanism configured to rotate the ramp ring relative to the upper body, the pre-load will be applied before rotationally locking the ramp ring and the upper body by way of the safety mechanism.

(95) Then, in one or more embodiments, a pressure-actuated release mechanism is used to secure the upper body to the lower body in an axial direction, with the ramp ring between a lower edge of the upper body and an upper edge of the lower body.

(96) Additionally, in one or more embodiments of the present disclosure, locking and rigidizing the seal assembly and casing hanger within a wellhead housing may include running an assembled seal assembly into the wellhead housing, landing the seal assembly on the casing hanger and sealing a seal of the seal assembly between the casing hanger and the wellhead housing. The seal assembly may then be locked to the wellhead housing. The process of locking the seal assembly may engage the locking mechanism into its locking profile within the wellhead housing. At the same time, the process of locking the locking mechanism may unlock a safety mechanism by compressing a spring-loaded rod into an elongated hole of the upper body and pushing a safety lock pin out of a pin counterbore of the upper body such that the ramp ring is no longer rotationally locked to the upper body. Then, the seal assembly may be lifted to preload the locking mechanism in place within the wellhead housing.

(97) In one or more embodiments, the space-out mechanism may then be actuated, taking up any axial gaps created by lifting on the seal assembly and rigidizing the seal assembly within the wellhead housing. Actuating the space-out mechanism may further include pressuring down an annulus between the casing hanger and a seal on the lower body to apply a downward force to the lower body, thereby shearing a pressure-actuated mechanism to enable the lower body to move downward with respect to the upper body. Moving the lower body in this manner causes the ramp ring to rotate, filling the gap between the lower body and the upper body. Once the space-out mechanism has been activated to rigidize the seal assembly and the casing body within the wellhead housing, the casing hanger seal may be seal tested to ensure that it is still properly sealing. Then, finally, the seal assembly may be released.

(98) The disclosure includes the following illustrative embodiments.

(99) A seal assembly, including: an upper body; a lower body having a seal for sealing between a first wellhead system component and a second wellhead system component; and a space-out mechanism disposed between the upper body and the lower body, the space-out mechanism including a ramp ring having at least one tapered surface, wherein the upper body has at least one tapered surface configured to interface with the at least one tapered surface of the ramp ring, wherein the ramp ring is configured to rotate relative to the upper body and the lower body, and wherein the at least one tapered surface of the ramp ring and the at least one tapered surface of the upper body are configured to bear against each other to rigidize the system.

(100) The seal assembly provided above may include any one or more of the following features: Feature 1: the tapered surfaces of the ramp ring and the upper body are complementary. Feature 2:

the taper of the at least one tapered surface of the ramp ring and the taper of the at least one tapered surface of the upper body have a slope between 0.5° and 7° . Feature 3: one or more of the ramp ring and the upper body have exactly three tapered surfaces. Feature 4: one or more pressure-actuated release mechanisms coupled between the upper body and the lower body. Feature 5: at least one of the one or more pressure-actuated release mechanisms is a shear pin. Feature 6: a spring coupled to the ramp ring and operable to circumferentially drive the ramp ring. Feature 7: a safety mechanism disposed at an interface of the upper body with the ramp ring, wherein the safety mechanism is configured to prevent rotation of the ramp ring with respect to the upper body until the safety mechanism is released.

(101) A system including: a wellhead; a casing hanger disposed at least partially in the wellhead; and a seal assembly, including: a space-out mechanism, the space-out mechanism including a ramp ring having at least one tapered surface; an upper body having at least one tapered surface configured to interface with the at least one tapered surface of the space-out mechanism; and a lower body having a seal for sealing between the casing hanger and the wellhead, wherein the ramp ring is configured to rotate relative to the upper body and the lower body, and wherein the at least one tapered surface of the ramp ring and the at least one tapered surface of the upper body are configured to bear against each other to rigidize the system.

(102) The system provided above may include any one or more of the following features: Feature 1: the tapered surfaces of the ramp ring and the upper body are complementary. Feature 2: the taper of the at least one tapered surface of the ramp ring and the taper of the at least one tapered surface of the upper body have a slope between 0.5° and 7° . Feature 3: one or more of the ramp ring and the upper body have three tapered surfaces. Feature 4: one or more pressure-actuated release mechanisms coupled between the upper body and the lower body. Feature 5: at least one of the one or more pressure-actuated release mechanisms is a shear pin. Feature 6: the wellhead is a subsea wellhead.

(103) A method, including: running a seal assembly into a wellhead housing until the seal assembly lands on a casing hanger; locking the seal assembly to the wellhead housing; lifting the seal assembly; and actuating a space-out mechanism of the seal assembly to rigidize the seal assembly within the wellhead housing, wherein the space-out mechanism includes a ramp ring having at least one tapered surface, and wherein an upper body of the seal assembly has at least one tapered surface configured to interface with the at least one tapered surface of the space-out mechanism.

(104) The method provided above may include any one or more of the following features: Feature 1: actuating the space-out mechanism expands at least a portion of the space-out mechanism to rigidize the tubing hanger system within the wellhead housing. Feature 2: actuating the space-out mechanism includes applying a pressure to the seal assembly to actuate one or more pressure-actuated release mechanisms. Feature 3: sealing a space between the wellhead housing and the casing hanger via a seal located on a lower body of the seal assembly. Feature 4: releasing a safety mechanism in response to movement of an actuator sleeve that locks the seal assembly to the wellhead housing, wherein the safety mechanism prevents rotation of the ramp ring with respect to the upper body until it is released.

(105) Although the present disclosure and its advantages have been described in detail, it should be understood that various changes, substitutions and alterations can be made herein without departing from the spirit and scope of the disclosure as defined by the following claims.

Claims

1. A hanger assembly, comprising: a main body; a lock ring coupled to the main body and configured to expand radially outward into a profile on an inner diameter of a wellhead; and an actuator, wherein the actuator is capable of transferring an axial force from the actuator into outward radial expansion of the lock ring toward the profile on the inner diameter of the wellhead;

a trigger rod coupled to the actuator and configured to unlock a space-out mechanism disposed between the main body and the lower body for actuation; wherein an interface between the actuator and the lock ring is shaped such that: axially downward movement of the actuator from a starting position to an intermediate position actuates the lock ring into the profile; and axially downward movement of the actuator from the intermediate position to a pre-load position applies a pre-load to the hanger assembly.

2. The hanger assembly of claim 1, wherein the actuator comprises a single tapered surface to interface with the lock ring.
3. The hanger assembly of claim 2, further comprising: an upper body coupled to the main body; and a split ring disposed along a radially inner surface of the upper body and configured to stop the axially downward movement of the actuator at the intermediate position.
4. The hanger assembly of claim 3, wherein the split ring is configured to retract into the radially inner surface of the upper body in response to additional axial force from the actuator to allow the axially downward movement of the actuator from the intermediate position to the pre-load position.
5. The hanger assembly of claim 2, wherein the single tapered surface is angled less than 4 degrees from vertical.
6. The hanger assembly of claim 5, wherein the single tapered surface is angled less than 2 degrees from vertical.
7. The hanger assembly of claim 1, wherein the actuator comprises two flat surfaces to interface with the lock ring, the two flat surfaces being separated by a step.
8. The hanger assembly of claim 1, wherein the actuator comprises a flat surface and a tapered surface to interface with the lock ring, the flat surface being located vertically below the tapered surface.
9. The hanger assembly of claim 1, further comprising: the lower body; and wherein the space-out mechanism comprising a ramp ring having at least one tapered surface.
10. The hanger assembly of claim 9, further comprising: a space-out mechanism release assembly configured to selectively release the space-out mechanism so that the space-out mechanism actuates; and wherein the trigger rod is configured to actuate the space-out mechanism release assembly upon the actuator moving to the intermediate position.
11. A method for installing a hanger assembly in a wellhead, comprising: positioning the hanger assembly into the wellhead, the hanger assembly comprising a main body, a lower body coupled to the main body, a lock ring coupled to the main body, and an actuator; moving the actuator axially downward with respect to the main body from a starting position to an intermediate position, wherein this movement of the actuator expands the lock ring radially outwardly into a profile on an inner diameter of the wellhead; unlocking a space-out mechanism disposed between the main body and the lower body so that the space-out mechanism actuates in response to the actuator moving from the starting position to the intermediate position, wherein a trigger rod is coupled to the actuator and configured to unlock the space-out mechanism upon the actuator moving to the intermediate position; actuating the space-out mechanism; and moving the actuator axially downward from the intermediate position to a pre-load position, wherein this movement of the actuator applies a pre-load to the hanger assembly.
12. The method of claim 11, further comprising: stopping the actuator at the intermediate position via a split ring disposed along a radially inner surface of an upper body coupled to the main body; retracting the split ring into the radially inner surface of the upper body in response to additional axial force from the actuator; and contacting a tapered surface of the actuator with the lock ring during movement of the actuator from the intermediate position to the pre-load position.
13. The method of claim 11, further comprising: contacting a first flat surface of the actuator with the lock ring during the movement of the actuator from the starting position to the intermediate position; and contacting a second flat surface of the actuator with the lock ring during the movement of the actuator from the intermediate position to the pre-load position, wherein the

second flat surface is separated from the first flat surface by a step.

14. The method of claim 11, further comprising: contacting a flat surface of the actuator with the lock ring during the movement of the actuator from the starting position to the intermediate position; and contacting a tapered surface of the actuator with the lock ring during the movement of the actuator from the intermediate position to the pre-load position, wherein the flat surface is located vertically below the tapered surface.

15. The method of claim 11, further comprising contacting a single tapered surface of the actuator with the lock ring during both the movement of the actuator from the starting position to the intermediate position and the movement of the actuator from the intermediate position to the pre-load position, wherein the single tapered surface is angled less than 4 degrees from vertical.

16. The method of claim 15, wherein the single tapered surface is angled less than 2 degrees from vertical.

17. The method of claim 11, wherein the space-out mechanism comprises a ramp ring having at least one tapered surface, and wherein actuating the space-out mechanism expands at least a portion of the space-out mechanism to rigidize the hanger assembly within the wellhead.

18. The method of claim 17, further comprising, after the actuator reaches the intermediate position, applying an overpull to lift the hanger assembly in a vertical direction relative to the wellhead to actuate the space-out mechanism.

19. The method of claim 17, further comprising maintaining the space-out mechanism in its actuated position during movement of the actuator from the intermediate position to the pre-load position.
